



# Whale shark

*Rhincodon typus*

Possibly the most misnamed aquatic creature in the world, the whale shark is not a whale, but a fish, although it resembles whales in terms of size. It belongs to the same class as sharks, skates, and rays. The word “shark” brings to mind a toothy, prey-crunching predator that could be a threat to humans, but the whale shark is in fact a gentle giant. This docile, slow-swimming filter feeder allows divers to grasp its large dorsal fin without displaying any signs of aggression. Ironically, the largest living fish in the world survives on a diet made up entirely of the ocean’s smallest organisms—tiny algae and animals known as phytoplankton and zooplankton.

## Filter feeding

Relatively little is known about the whale shark’s life and behavior. The average adult can be up to 39 ft (12 m) in length—although there are unverified reports of specimens as long as 75 ft (23 m)—and weighs around 10 tons (9 metric tons). The whale shark’s huge, flat head ends in a mouth that is almost as wide as its body and contains hundreds of minuscule teeth, the function of which is unknown. It has specialized, sievelike filter pads lining the gill arches, which separate food from seawater. Unusually for a shark, its mouth is at the end of its snout rather than underneath. As well as non- or slow-moving microscopic organisms, whale sharks also eat small fish, tiny squid, fish eggs, and larvae—in short,

anything small enough to flow in with water but large enough to be trapped by their filter pads. Mostly solitary, whale sharks are occasionally seen in loosely organized schools of up to 100 individuals where food is plentiful.

Whale sharks migrate thousands of miles through the world’s oceans. Individuals can be tracked using satellite tags and can be identified by the pattern of spots on their bodies—no two patterns are alike, just like human fingerprints. Adult males can be distinguished from females by the presence of “claspers”—external protrusions on their undersides that channel sperm into the female during mating. Little is known about when and how whale sharks breed, but a female retains up to 300 eggs inside her body until they hatch—a characteristic known as ovoviviparity. She then gives birth to live young or “pups,” although it is believed that not all the pups are born at the same time.

## Uncertain future

Once it has reached sexual maturity, which is at about 30 years of age, the whale shark’s primary predator is man. They are hunted for food supplements, such as shark liver oil; their fins, which are made into soup; their meat; and their skin, which is made into leather. However, many adult whale sharks carry scars that suggest they were attacked, possibly by orcas or other sharks, at some point in their lives. It is believed that whale sharks can live up to 70–100 years in the wild.



### △ SPECIALIZED GILLS

Cartilage rods support the spongy filter pads, which trap food such as plankton and tiny fish and squid.

